Notes on Computational Fluid Dynamics: General Principles

2.8 Flow in a volume

We presented conservation of momentum, Eq. (2.19), in terms of the material derivative $D\mathbf{u}/Dt$. To solve the equation in this form, would require a method that tracks particles of fluid as they move around.

Generally it is easier to solve the equation by *fixing* the volume of space and solving for the fluid motion through it. To enable this, we replace the material derivative for $\mathbf{u}$ using the following expression:

$$\boxed{\rho\frac{D\mathbf{u}}{Dt} = \frac{\partial\rho\mathbf{u}}{\partial t} + \nabla\cdot(\rho\mathbf{u}\mathbf{u})} \tag{2.26}$$

Eq. (2.26) is derived from the material derivative Eq. (2.14) by

$$\rho\frac{D\mathbf{u}}{Dt} = \rho\frac{\partial\mathbf{u}}{\partial t} + \rho\mathbf{u}\cdot\nabla\mathbf{u}$$

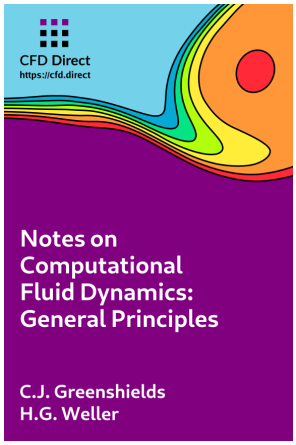
$$\underbrace{\frac{\partial\rho\mathbf{u}}{\partial t} - \mathbf{u}\cancel{\frac{\partial\rho}{\partial t}}}_{=0 \text{ by Eq. (2.8)}} + \nabla\cdot(\rho\mathbf{u}\mathbf{u}) - \mathbf{u}\cancel{\nabla\cdot(\rho\mathbf{u})}$$

Conservation of momentum can then be written

$$\frac{\partial\rho\mathbf{u}}{\partial t} + \nabla\cdot(\rho\mathbf{u}\mathbf{u}) = \nabla\cdot\boldsymbol{\sigma} + \rho\mathbf{b}. \tag{2.27}$$

A term of the form $\nabla\cdot(\mathbf{u}\Psi)$ or $\nabla\cdot(\rho\mathbf{u}\Psi)$ represents the bulk motion of the fluid at velocity $\mathbf{u}$, which transports the property Ψ (a tensor of any rank) by advection[6]. For example, if the property Ψ is temperature, then advection will transport heat from one region of the flow domain to another.

Advection contains a divergence derivative so represents a flux across a surface per unit volume. The example in Eq. (2.27) represents the flux of



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momentum, where the advected property is itself $\mathbf{u}$ (or $\rho\mathbf{u}$, depending whether ρ is associated with bulk flow or the advected property).

Outer product of two vectors

The advection term in Eq. (2.27) includes a product of two vectors $\mathbf{uu}$. This is the outer product of two vectors[7] which produces a tensor by

$$\mathbf{ab} = \begin{pmatrix} a_x b_x & a_x b_y & a_x b_z \\ a_y b_x & a_y b_y & a_y b_z \\ a_z b_x & a_z b_y & a_z b_z \end{pmatrix}. \tag{2.28}$$

Inner product of two tensors

The inner product of two tensors, *e.g.* $\mathbf{T} \cdot \mathbf{S}$ produces a tensor $\mathbf{Y}$ where the components are (replacing i, j with x, y, z)

$$Y_{ij} = T_{ix} S_{xj} + T_{iy} S_{yj} + T_{iz} S_{zj}. \tag{2.29}$$

Identity tensor

The identity tensor $\mathbf{I}$ is the tensor equivalent of unity (one) such that for any tensor $\mathbf{T} = \mathbf{I} \cdot \mathbf{T} = \mathbf{T} \cdot \mathbf{I}$,

$$\mathbf{I} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{2.30}$$

[6] The term "convection" is sometimes used, but it has conflicting meanings, so we avoid it here.

[7] sometimes written "$\mathbf{a} \otimes \mathbf{b}$", but we write it as a product with no symbol, similar to a scalar multiplication.

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